

# FINAL EXPLANATION: CORE MATHEMATICS GRADE 10

Teacher Key: Exemplar Answers and Marking Guide

## FINAL EXPLANATION: CORE MATHEMATICS GRADE 10: SUB-STRAND 2.7 SURFACE AREA AND VOLUME OF SOLIDS – FINAL EXPLANATION

Teacher Key: Exemplar Answers (do not give to students)

### INSTRUCTIONS GIVEN TO STUDENTS

Driving question: Why does the same amount of metal sheet make containers that hold very different amounts of grain, and how can a Kenyan artisan use mathematics to design the best solid shape for storage or construction? Use what you learned in Lessons 1 to 8 to answer each question below. Write in full sentences, show every step of your calculations, include units, and refer to the posho mill in Eldoret where you can. Round answers to the accuracy asked for. Use your Driving Question Board and your cumulative model portfolio to help you.

### Part 1: What the metal sheet really tells us

**A posho mill owner in Eldoret has a rectangular metal box for storing maize. It measures  $l = 120$  cm,  $w = 80$  cm and  $h = 100$  cm. The fundi who made it used the same total area of iron sheet for a cylindrical drum. (a) Write down the formula you use and calculate the total area of iron sheet used for the box. (b) Calculate the volume of the box in  $\text{cm}^3$  and in  $\text{m}^3$ . (c) In your own words, explain what surface area measures and what volume measures. Say why knowing one of them does not tell you the other.**

(a) The box is a rectangular prism, so  $SA = 2(lw + lh + wh) = 2(120 \times 80 + 120 \times 100 + 80 \times 100) = 2(9,600 + 12,000 + 8,000) = 2 \times 29,600 = 59,200 \text{ cm}^2$ . This is the total area of iron sheet the fundi used, and the same area was used for the drum. (b)  $V = l \times w \times h = 120 \times 80 \times 100 = 960,000 \text{ cm}^3$ . Since  $1 \text{ m}^3 = 1,000,000 \text{ cm}^3$ ,  $V = 0.96 \text{ m}^3$ . (c) Surface area is the total area of all the faces on the outside of a solid. It tells us how much material, such as iron sheet, covers the solid. Volume is the amount of space inside the solid, so it tells us how much maize can fit. They are different properties. Two containers can use exactly the same area of iron sheet but be shaped differently, so they can enclose different amounts of space. This is why equal surface area does not mean equal volume.

### Part 2: Solving the posho mill mystery with numbers

**The cylindrical drum is closed at top and bottom, is 100 cm tall, and is made from the same  $59,200 \text{ cm}^2$  of iron sheet as the**

(a)  $TSA = 2\pi r^2 + 2\pi rh$ . With  $r = 59.2$  cm and  $h = 100$  cm:  $2\pi(59.2)^2 = 2\pi(3,504.64) \approx 22,020 \text{ cm}^2$ , and  $2\pi(59.2)(100) \approx 37,200 \text{ cm}^2$ . The total is about  $59,200 \text{ cm}^2$ , which matches the iron sheet used for the box. (b)  $V = \pi r^2 h = \pi \times 3,504.64 \times 100 \approx 1,100,000 \text{ cm}^3$ , which is about  $1.10 \text{ m}^3$ . (c) The box holds  $960,000 \text{ cm}^3$  ( $0.96 \text{ m}^3$ ). The drum holds about  $1,100,000 \text{ cm}^3$ , so it holds more maize. The difference is about  $140,000 \text{ cm}^3$ , which is about 140 litres. As a percentage of the box's volume this is  $140,000 \div 960,000 \times$

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| <p>box. Its radius works out to about 59.2 cm. (a) Show, using the formula for the total surface area of a closed cylinder, that this radius matches the sheet area given. A small rounding difference is acceptable. (b) Calculate the volume of the drum in <math>\text{cm}^3</math> and in <math>\text{m}^3</math> to 2 decimal places. (c) Compare it with the box from Part 1. State which holds more maize, by how many <math>\text{cm}^3</math>, and by about what percentage of the box's volume.</p> | <p><math>100 \approx 15\%</math>. So the drum holds about 15% more grain from the same sheet of iron. This is the answer to the mill owner's puzzle.</p> |
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| Part 3: Why the cylinder wins                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
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| <p>Use your understanding of shapes to explain to the mill owner why the cylindrical drum holds more maize than the rectangular box even though both used the same area of iron sheet. In your explanation: (a) say what the same-height drum and box have in common and what is different about their base shapes; (b) use the idea of the isoperimetric principle from Lesson 8; (c) explain why this means your answer is not just a coincidence of these particular dimensions.</p> | <p>(a) Both containers are prisms-like solids that are 100 cm tall and use the same total area of iron sheet. The difference is the shape of the cross-section, which is a circle for the drum and a rectangle for the box. For a solid with the same height, volume = base area <math>\times</math> height, so the container with the larger base area holds more. (b) The isoperimetric principle says that, for a given perimeter, a circle encloses a greater area than any other closed shape, including a rectangle. The circular base of the drum therefore encloses more area for the same amount of boundary material. A larger base area, multiplied by the same height, gives a larger volume. (c) The result follows from the geometry of the shapes, not from the particular numbers. Among solids with equal surface area, the cylinder encloses a greater volume than a rectangular prism of the same height. This is why the difference in capacity is a result of shape, not of the amount of sheet metal.</p> |

| Part 4: Choosing the right formula for the fundi                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
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| <p>A metalsmith makes a conical grain funnel with base radius <math>r = 30</math> cm and vertical height <math>h = 40</math> cm. (a) Calculate the slant height. Explain why the slant height, not the vertical height, is used to find the sheet metal</p> | <p>(a) The slant height <math>l = \sqrt{r^2 + h^2} = \sqrt{30^2 + 40^2} = \sqrt{900 + 1,600} = \sqrt{2,500} = 50</math> cm. The slant height is used because it lies along the outer surface of the cone from the apex to the base edge. This is the length the metalsmith actually cuts in the sheet, which unrolls into a sector of a circle with radius <math>l</math>. The vertical height is inside the cone and is not on the surface. (b) <math>\text{CSA} = \pi r l = \pi \times 30 \times 50 = 1,500\pi \approx 4,712 \text{ cm}^2</math>. (c) <math>V = \frac{1}{3}\pi r^2 h = \frac{1}{3} \times \pi \times 900 \times 40 = 12,000\pi \approx 37,699 \text{ cm}^3</math>. (d) A cylinder of the same radius and height has <math>V = \pi r^2 h = 36,000\pi \text{ cm}^3</math>. The cone is one third of this. We saw this in the water-pouring experiment, where exactly three full cones of water filled the cylinder.</p> |

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| <p>needed. (b) Find the curved surface area of the funnel, leaving your answer in terms of <math>\pi</math> and then to the nearest <math>\text{cm}^2</math>. (c) Find the volume of the cone in terms of <math>\pi</math> and then to the nearest <math>\text{cm}^3</math>. (d) State how the volume of this cone compares with a cylinder of the same base radius and height, and how you know.</p> |  |
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| Part 5: Advice for a Kenyan artisan                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
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| <p>A jua kali metalsmith in Kamukunji has a fixed amount of iron sheet and wants to build a grain store that holds as much maize as possible. Write a short piece of advice to the metalsmith that answers the driving question. Your advice must: (a) explain the difference between surface area and volume and say how the metalsmith can calculate each for a cylinder and for a cuboid; (b) recommend a shape from the Eldoret example and justify it with evidence from your calculations; (c) explain what the metalsmith should do for a composite solid such as a cylinder topped with a cone or hemisphere, including how to treat surfaces where two parts join.</p> | <p>Dear fundi, surface area is the amount of iron sheet that covers a solid, and volume is the amount of maize it can hold. These are different, so you must calculate both before you cut any metal. For a cylinder, <math>\text{TSA} = 2\pi r^2 + 2\pi rh</math> and <math>V = \pi r^2 h</math>. For a cuboid, <math>\text{TSA} = 2(lw + lh + wh)</math> and <math>V = lwh</math>. In Eldoret, a rectangular box of <math>120 \text{ cm} \times 80 \text{ cm} \times 100 \text{ cm}</math> and a cylindrical drum <math>100 \text{ cm}</math> tall were made from the same <math>59,200 \text{ cm}^2</math> of iron sheet. The box holds <math>960,000 \text{ cm}^3</math> (<math>0.96 \text{ m}^3</math>), while the drum holds about <math>1,100,000 \text{ cm}^3</math> (<math>1.10 \text{ m}^3</math>). That is about 15% more maize from the same metal. So if you want the most storage from a fixed amount of sheet, choose a cylinder rather than a rectangular box. This works because a circle encloses more area than a rectangle for the same perimeter. If you build a composite shape, such as a cylinder with a cone roof or a hemisphere top, split it into its parts. Add the volumes of all the parts to get the total volume. For the surface area, add only the exposed surfaces and leave out the internal circle where the two parts are joined, because no iron sheet is needed there. Check each design by calculating its surface area and volume so that your decisions are based on mathematics and not on appearance.</p> |

| RUBRIC                                                           |                                                                                 |                                                                                 |                                                                                   |
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| Criterion                                                        | Excellent (4)                                                                   | Proficient (3)                                                                  | Developing (1–2)                                                                  |
| Correct use of surface area and volume formulae and calculations | Chooses the right formula every time, shows all steps, gets correct values with | Uses mostly correct formulae and gets most values right, with minor arithmetic, | Uses wrong formulae or leaves out steps, with several incorrect values or missing |

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|                                                                  | units and sensible rounding for the box, drum and cone.                                                                                                           | rounding or unit slips.                                                                              | units.                                                                                       |
| Distinguishing surface area from volume                          | Clearly explains that surface area is the material covering a solid and volume is the space inside, and uses the Eldoret containers to show they are independent. | States the difference correctly but gives a limited link to the phenomenon.                          | Confuses surface area and volume or treats equal surface area as meaning equal volume.       |
| Explaining why the cylinder holds more (isoperimetric reasoning) | Links the circular cross-section to a greater base area for the same perimeter and explains that this applies generally, not just to these numbers.               | Says the circle is more efficient and links this to volume, but the reasoning is incomplete.         | Gives no mathematical reason, or only says the cylinder is bigger.                           |
| Use of evidence and Kenyan context in the design advice          | Uses specific data from the Eldoret example, handles slant height and composite solids correctly, and gives practical, justified advice.                          | Uses some evidence and gives reasonable advice, with small gaps on slant height or composite solids. | Gives advice with little or no evidence, or omits the composite solid or slant height ideas. |

## ATTRIBUTION AND LICENSING

**Curriculum Alignment:** Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

**Content & Pedagogy:** Developed and adapted by the educator team at SeaVuria (registered 501(c)(3) in the United States and registered NGO in Kenya, No. 872-850A-BF11), who designed the pedagogical framework, lesson templates and exemplar lessons on which this material is based.

**Production & Layout:** Educational design and curriculum sequencing by SeaVuria. Technical layout, structural formatting, and digital optimization/implementation executed in partnership with ARES Education (registered 501(c)(3) in the United States and Canada (as Afretech Aid) and registered NGO in Kenya), with lesson drafting and compilation assisted by AI (Claude, Anthropic).

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